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A Comprehensive Guide to Latent Dirichlet Allocation

Building a Solid Foundation with Key Statistical Concepts

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Abstract

To fully comprehend Latent Dirichlet Allocation (LDA) and its applications, it is essential to build a strong foundation in the underlying statistical principles. This guide offers an in-depth exploration of LDA, explaining both its mechanics and the key theoretical frameworks that inform its design. By covering topics such as probability distributions, Bayesian inference, and Dirichlet priors, we aim to equip readers with the knowledge needed to understand LDA from first principles. Section 1 introduces the concept of topic modeling with a focus on LDA. Section 2 provides an overview of the probability distributions integral to LDA. Section 3 delves into Bayesian inference, particularly Gibbs sampling and its significance. Section 4 explores word representation and processing within the model. Section 5 synthesizes these concepts, presenting a detailed explanation of LDA's terminology, the implementation of a collapsed Gibbs sampler, expected outcomes, and a brief overview of potential applications.

***Key words*:** Latent Dirichlet Allocation, Topic modeling, Bayesian inference, Statistical foundations, Dirichlet priors, Generative models, Gibbs sampling..

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1. Introduction

In the expanding field of topic modelling, numerous articles and multimedia materials focus on explaining Latent Dirichlet Allocation (LDA). While both approaches offer some insight into what LDA achieves, they often miss a balanced explanation that is both accessible and in-depth. Highly theoretical resources can be challenging for readers seeking a clearer, more intuitive understanding, while high-level summaries, although easier to follow, may lack the depth needed to truly understand how LDA operates.

The purpose of this guide is to bridge this gap by providing a straightforward yet thorough explanation of LDA. We aim not only to describe what LDA is but also to present, from the ground up, the fundamental statistical concepts necessary for a comprehensive understanding. This includes key concepts that are essential to grasp before diving into the mechanics of LDA.

In this text, we will cover the core principles behind LDA, along with an introductory path through the relevant statistical distributions, setting a solid foundation before introducing LDA as a generative model.

1.1. Topic Modelling and LDA

Topic modelling as a statistical technique used to uncover hidden topics within a collection of documents. By grouping words that frequently co-occur, it helps discover patterns and relationships within text data. Typically, topic modeling is unsupervised, meaning it does not require pre-labeled because it doesn't require pre-labeled data. Instead, it infers topics based on the across documents, with the corpus serving as the primary input. Topic modeling's strength lies in its ability to analyze vast amounts of text efficiently, revealing insights and trends without the need to read each document individually.

Some of the most common algorithms in topic modeling include Latent Dirichlet Allocation (LDA), Latent Semantic Analysis (LSA), and Nonnegative Matrix Factorization (NMF). This article focuses on LDA, a widely used generative probabilistic model for collections of discrete data developed by Blei et al. 2003. Researchers frequently use LDA for topic modeling due to its effectiveness in identifying meaningful topics within large text corpora Vayansky I (2020).

The primary objectives of LDA are to:

- **Identify Topics:** Determine the latent topics that best explain patterns of word co-occurrence across the documents.
- **Assign Topic Distributions:** Provide each document with a distribution over topics indicating the contribution of each topic to the document.
- **Infer Word Distributions:** Estimate word distributions for each topic, identifying key terms that define each topic.

A critical concept in the LDA model is how documents and topics can be represented probabilistically. Figure 1 illustrates the concept of LDA in topic modeling. In this example, the vocabulary of the corpus has three words, and the model is set to three topics. Each topic is represented as a distribution over the vocabulary, demonstrating how LDA allows documents to be expressed as mixtures of topics.

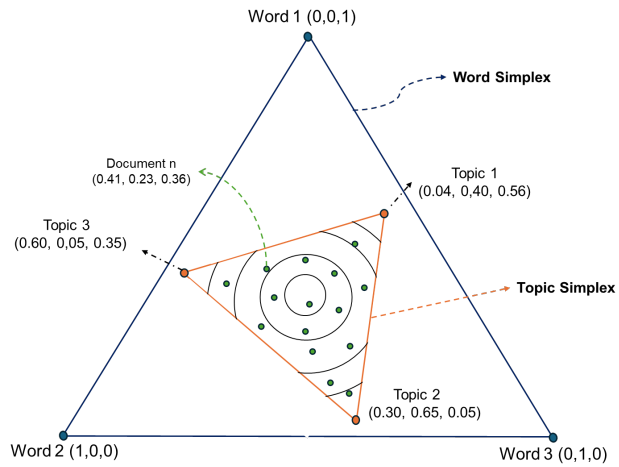

FIGURE 1: Representation of document, word and topics in both word and topic simplex. Source: Authors

The probabilistic framework provided by LDA is a significant advantage, as it mirrors the complexity of real-world documents where multiple themes often coexist within a single document. For example, a news article might discuss both politics and sports, and a word like "policy" could appear across multiple topics. LDA effectively captures this nuance by assigning probabilities rather than categorizing each word and document within a single topic.

2. Important probability distributions

In the previous section, we introduced Latent Dirichlet Allocation (LDA) and its primary goal of discovering hidden topics in large text datasets. To fully understand how LDA functions, we now need to explore the key probability distributions underlying the model: the Poisson distribution, which can model word occurrence frequencies; the multinomial distribution, governing word distributions within topics; and the Dirichlet distribution, which controls the mixture of topics in each document. Additionally, the Beta distribution, a specific case of the Dirichlet distribution, is useful for modeling binary outcomes.

This section focuses on these essential distributions, explaining their roles in LDA's structure. Understanding these distributions provides a clear basis for

exploring the generative model behind LDA, illustrating how LDA uses these statistical frameworks to assign topics to words and documents.

2.1. Poisson Distribution

The Poisson distribution is a discrete probability distribution suitable the number of events occurring within a fixed interval $[0, t]$.

A random variable X is said to follow a Poisson distribution with parameter $\lambda > 0$, denoted as $X \sim P(\lambda)$, if its probability mass function is given by:

$$p(x;\lambda) := f_X(x) = \begin{cases} \frac{e^{-\lambda}}{x!}\lambda^x & \text{, if } x = 0,1,2,... \\ 0 & \text{, Otherwise} \end{cases}$$

Similarly, its cumulative distribution function F is given by:

$$p(t;\lambda) := F(t) = P(X \leq t) = \sum_{k \leq t} p(x;\lambda)$$

The Poisson distribution is appropriate when:

- k is the number of occurrences within an interval with possible values $0, 1, 2, ...$
- Events occur independently of each other.
- The event rate is constant over t.
- Two events cannot occur simultaneously.

In LDA, the Poisson distribution can model the frequency of word occurrences within a document. Since LDA assumes that each document is a mixture of topics and each topic generates words with certain probabilities, the Poisson distribution captures the discrete nature of word counts in documents, supporting the model's generative process. (For further properties and proofs of the Poisson distribution, refer to the appendix.)

2.2. Beta Distribution

The Beta distribution is a continuous probability distribution defined in the interval $[0, 1]$ and parameterized by two shape parameters, $\alpha > 0$ and $\beta > 0$. The generalization of this distribution to multiple variables is known as the Dirichlet distribution (which will be discussed in more detail later in this document).

If X is a continuous random variable with beta distribution, we write $X \sim B(\alpha, \beta)$ and its probability density function:

$$p(x;\alpha,\beta) := f_X(x) = \begin{cases} \frac{1}{B(\alpha,\beta)} \cdot x^{\alpha-1}(1-x)^{\beta-1} & \text{, if } x \in (0,1) \quad \text{where } \alpha, \beta > 0 \\ 0 & \text{, otherwise} \end{cases}$$

Where $B(\alpha, \beta)$ is the beta function:

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\, dx = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

In LDA, each topic is represented by a distribution over words, often modeled using a Dirichlet distribution—a multivariate generalization of the Beta distribution. The Beta distribution helps in modeling probabilities in binary scenarios, such as the presence or absence of a word. (For additional properties and proofs, see the appendix.)

2.3. Multinomial Distribution

The multinomial distribution is a multivariate extension of the binomial distribution, suitable for modeling outcomes with more than two possibilities. If an experiment has n trials with k outcomes and $X = [X_1, X_2, \ldots, X_k]$ denotes the counts of each outcome, then X follows a multinomial distribution.

Let X be a discrete random vector of size $k \times 1$ and $n \in \mathbb{N}$ such that:

$$R_X = \left\{ x \in \{0, 1, 2, \ldots, n\}^k : \sum_{i=1}^{k} x_i = n \right\}$$

and let $P = \{p_1, p_2, \ldots, p_k\}$ be a probability vector of size $k \times 1$ such that $\sum_{i=1}^{k} p_i = 1$. We can say that X follows a multinomial distribution with probabilities P and number of trials n if its probability function is given by:

$$p(x_1, x_2, \ldots, x_k; n, p_1, p_2, \ldots, p_k) = \begin{cases} \binom{n}{x_1, x_2, \ldots, x_k} \prod_{i=1}^{k} p_i^{x_i} & , \text{ if } x \in R_X \\ 0 & , \text{ otherwise} \end{cases}$$

Here $\binom{n}{x_1, x_2, \ldots, x_k}$ is the multinomial coefficient $\frac{n!}{x_1! x_2! \cdots x_k!}$.

In LDA, the multinomial distribution models the relationship between topics and words. Each document is generated from a mixture of topics, with each topic represented by a multinomial distribution over a vocabulary. This distribution allows us to interpret topics in terms of the words most likely to appear in them, aiding in the model's interpretability.(For more details, consult the appendix.)

2.4. Dirichlet Distribution

The Dirichlet distribution is a continuous multivariate probability distribution often used to model uncertainty over a probability vector. It generalizes the Beta distribution to multiple variables, making it ideal for scenarios where multiple mutually exclusive events need probability assignments. In LDA, the Dirichlet

distribution models the uncertainty of topic proportions in documents, helping to ensure valid probabilities that sum to one. Let X denote the probability of an event. If this probability is unknown, we can treat X as a random variable and assign it a Beta distribution with parameters α and β.

However, if X is a *vector of unknown probabilities* of mutually exclusive events, it can be considered as a random vector and assigned a Dirichlet distribution. Let X be a continuous random vector of size $K \times 1$ where:

$$R_X = \left\{ x \in \mathbb{R}^K : \forall k, x_k \geq 0 \text{ and } \sum_{k=1}^{K} x_k \leq 1 \right\}$$

It is said that X follows a Dirichlet distribution with parameters $\alpha = (\alpha_1, \ldots, \alpha_K, \alpha_{K+1})$ if and only if its probability function is given by:

$$f_X(x) = \begin{cases} c \left(\prod_{k=1}^{K} x_k^{\alpha_k - 1} \right) \left(1 - \sum_{k=1}^{K} x_k \right)^{\alpha_{k+1} - 1} & , \text{ if } x \in R_X \\ 0 & , \text{ if } x \notin R_X \end{cases}$$

where the normalization constant c is the multinomial beta function, which can be expressed in terms of the gamma function:

$$c = \frac{1}{B(\alpha)} = \frac{\Gamma\left(\sum_{k=1}^{K+1} \alpha_k \right)}{\prod_{k=1}^{K+1} \Gamma(\alpha_k)}$$

The simplex of a Dirichlet distribution is a geometric representation of all possible combinations of probabilities for a vector of K mutually exclusive events. It exists in $K - 1$ dimensions. for $K = 2$, the simplex is a line segment; for $K = 3$, it is a triangle; and for $K = 4$, it forms a tetrahedron. Each point within the simplex represents a valid probability distribution where all probabilities are nonnegative and sum to one. The vertices of the simplex correspond to extreme distributions, where one event has a probability of 1 and all others a probability of 0.

Figure 2 depicts a 3D simplex (a triangle in 3D space for a Dirichlet distribution with three variables (x, y, z). Each vertex represents a point where one of the three variables takes the value 1, while the others are 0. Every point within the triangle corresponds to a different composition of these variables, with values that sum to 1, which is characteristic of a Dirichlet distribution. The coordinates of each point indicate the proportion of each variable in that composition. For example, the point (0.72, 0.14, 0.14) represents a composition with 72% in the x dimension, 14% in the y dimension and 14% in the z dimension. This simplex provides a visualization of the possible outcomes of a Dirichlet distribution over three categories.

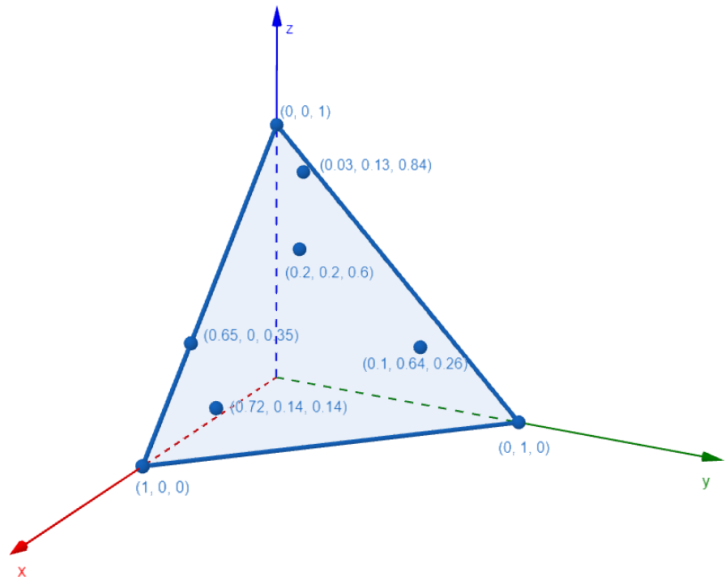

FIGURE 2: 3D simplex representation of points drawn from a Dirichlet distribution, illustrating different compositions of three variables (x, y, z) summing to 1. Source: Authors.

The parameters $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_K)$ influence the shape of the simplex, with higher values for a parameter indicating a greater likelihood of the corresponding event. This representation is essential for understanding how the Dirichlet distribution models uncertainty over event probabilities. It is particularly useful in applications like Latent Dirichlet Allocation (LDA), where it helps infer topic distributions within documents.

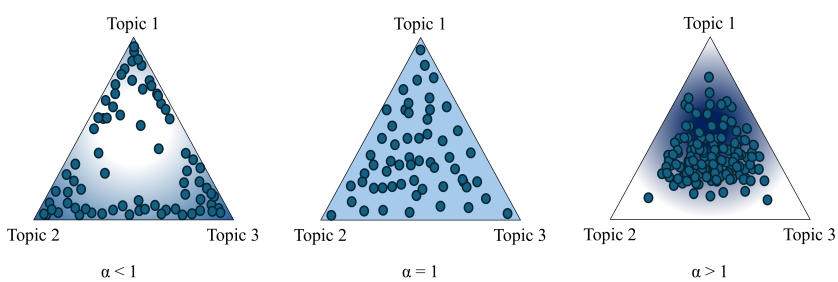

FIGURE 3: Simplex visualization of Dirichlet-distributed topic proportions in documents for given α values. Source: Authors.

As shown in figure 3, each point on the simplex represents a document. On the left (where $\alpha < 1$) the topic distributions are sparse, with most documents dominated by a single topic. In the middle (where $\alpha = 1$), topics are more evenly distributed across documents, creating a balanced mix. On the right (where $\alpha > 1$) documents show an even distribution of topics, meaning each document includes

a broader mix of topics. When $\alpha < 1$ the Dirichlet distribution tends to favor sparse distributions, where only a few topics have significant probabilities in each document while others are close to zero. This sparsity is useful in Latent Dirichlet Allocation (LDA) because we generally expect each document to cover a small subset of possible topics, making the topic model more interpretable.

The Dirichlet distribution plays a key role in LDA by providing a prior for topic proportions across documents. Each document is assumed is generated from a mixture of topics, and the Dirichlet distribution models the uncertainty in the share of each topics within a document, ensuring that topic proportions remain valid probabilities (summing to one). This allows LDA to capture the variability of topic distributions across documents and effectively uncover underlying topic structures in a text corpus, making it a powerful tool for topic modeling.

For more details on the properties of the Dirichlet distribution, including its covariance matrix and cross-moments, see the appendix.

3. Preliminaries of Bayesian and MCMC methods

3.1. Bayesian Inference

Bayesian inference is a statistical method that updates the probability of a hypothesis as more evidence or data becomes available. In essence, it involves learning about the parameters $\theta = (\theta_1, ..., \theta_k)$ of a population based on a dataset $x = (x_1, ..., x_n)$. The specific model definition depends on expert judgment, as there is no universally fixed approach.

Unlike frequentist methods, which view parameters as fixed values and rely on long-run frequencies, Bayesian inference treats parameters as random variables and interprets uncertainty in probabilistic terms (Box & Tiao, 1992). This makes Bayesian inference particularly powerful in situations where prior information is available for complex models involving latent (unobserved) variables.

Bayesian inference is built on Bayes' theorem, which comprises three main components: the prior, the likelihood and the posterior. Bayes' theorem provides a framework for updating our beliefs in light of new data. It can be expressed as:

$$P(\text{belief} \mid \text{data}) = \frac{P(\text{belief}) \cdot P(\text{data} \mid \text{belief})}{P(\text{data})}$$

The components of Bayes' theorem are as follows:

- $P(\text{belief})$ or $P(\theta)$: This is the prior, representing is our initial belief about a hypothesis before seeing any data. It reflects our prior knowledge or assumptions about the probability of the hypothesis.
- $P(\text{data} \mid \text{belief})$ or $P(x|\theta)$: This is the likelihood, the probability of observing the data given that the hypothesis (or belief) is true. It shows how likely the observed data is if our belief is correct and characterizes how x is generated given θ.

- $P(\text{belief} \mid \text{data})$ or $P(\theta|x)$: This is the posterior, representing the updated belief after considering the new data. It is the probability of the hypothesis being true given the observed data.

The term $P(data)$ or $P(x)$ is the total probability of the data occurring under all possible hypotheses, also known as a normalization term. It ensures that posterior probabilities up to 1. In Bayesian inference, rather than calculating this term directly (which can be challenging for complex models), we often work with proportional values, expressing the posterior as Geyer 1994:

$$P(\theta|x) \propto P(x|\theta) \cdot P(\theta)$$

Formally, suppose we have a random sample $X_1, \ldots, X_n \sim f(x|\theta)$ where $\theta \in H$ is unknown.

We aim to find a probability distribution of θ:

$$p(\theta|x_1, \ldots, x_n) = \frac{p(x_1, \ldots, x_n|\theta)p(\theta)}{c}$$

Where c is the normalization term given by:

$$c = \int_H p(x_1, \ldots, x_n|\theta)p(\theta)d\theta$$

The likelihood can also be expressed as:

$$p(x_1, \ldots, x_n|\theta) = \prod_{k=1}^{n} f(x_k|\theta) \equiv L(\theta)$$

Here, $p(\theta)$ is the prior distribution, reflecting information outside the sample, such as expert judgment. The posterior then becomes:

$$p(\theta|X) = \frac{L(\theta|X)p(\theta)}{c} \propto L(\theta|X)p(\theta)$$

(For an example with a binomial model, please see the appendix.)

Bayesian inference is vital in Latent Dirichlet Allocation (LDA) because it provides a framework for estimating the hidden structure within data, particularly topic distributions over documents and word distributions over topics (Anupriya and Karpagavalli 2015). LDA's goal is to latent topics that generate observed words in a corpus. Through Bayesian inference, we compute the posterior distribution of these hidden variables (topics) given the observed data (words in documents). By treating topic and word distributions as random variables with Dirichlet priors, Bayesian inference allows us to update our beliefs about the parameters of these distributions as we observe more data. Techniques like Gibbs sampling help approximate the posterior distribution efficiently, making it feasible to apply LDA to large corpora.

3.2. Conjugate families of distributions

In Bayesian statistics, the concept of *conjugate priors* refers to a special relationship between prior and posterior distributions. A family of prior distributions is called *conjugate* to a likelihood function if the posterior distribution belongs to the same family as the prior.

Formally, if we have a likelihood function for data x given a parameter θ of the form $P(x|\theta)$ and a prior distribution $P(\theta)$ that is conjugate to the likelihood, then the posterior distribution $P(\theta|x)$ retains the same functional form as the prior $P(\theta)$.

Some common examples of conjugate families include:

- **Beta distribution** as a conjugate prior for the **Binomial likelihood**,
- **Gamma distribution** as a conjugate prior for the **Poisson likelihood**,
- **Gamma distribution** as a conjugate prior for the **Exponential likelihood**,
- **Dirichlet distribution** as a conjugate prior for the **Multinomial likelihood**.

The Dirichlet-Multinomial model will be explained in Section 3.1.1. For further illustration, see the Appendix for an example of the Gamma-Poisson model.

The primary advantage of using conjugate priors is that they simplify the calculation of the posterior distribution. This leads to analytical tractability, allowing us to avoid complex integrations (Gutiérrez-Peña E. and Bernardo 1997). When a conjugate prior is chosen, the posterior can be computed in closed form, making it easier to perform inference and update beliefs as new data arrives.

3.2.1. Dirichlet as conjugate prior multinomial

The Dirichlet distribution is a multivariate distribution for a set of quantities $p_i \geq 0$ that sum to 1, making it ideal for modeling a set of probabilities of mutually exclusive events (a partition).

If we have an experiment that can be modeled as a multinomial and we want to estimate p_i we can use either the maximum likelihood estimators (frequentist approach) or the Bayesian approach. In the Bayesian approach, we start with a prior distribution for each p_i, which is modeled by the Dirichlet distribution. The prior for each p_i is defined by the choice of parameters α_i in the Dirichlet distribution.

The Bayesian posterior distribution for p given the data x is:

$$P(p|x) \propto P(x|p) \cdot P(p)$$

where $P(x|p)$ is the multinomial likelihood and $P(p)$ is the Dirichlet prior. Multiplying the multinomial probability density function by the Dirichlet prior results in:

$$P(p|x) \propto \frac{n!}{x_1!x_2!\dots x_k!} \prod_{i=1}^{k} p_i^{x_i} \cdot \frac{\Gamma\left(\sum_{i=1}^{k} \alpha_i\right)}{\prod_{i=1}^{k} \Gamma(\alpha_i)} \prod_{i=1}^{k} p_i^{(\alpha_i - 1)}$$

$$P(p|x) \propto \prod_{i=1}^{k} p_i^{\alpha_i + x_i - 1}$$

The posterior distribution, up to a constant, has the form of a Dirichlet distribution with parameters $\alpha_i + x_i$. Since the posterior is of the same family as the prior (Dirichlet), the Dirichlet is indeed a conjugate prior for the multinomial distribution. We can add the normalization term to obtain the final posterior:

$$P(p|x) \propto \frac{\Gamma\left(\sum_{i=1}^{k} \alpha_i + x_i\right)}{\prod_{i=1}^{k} \Gamma(\alpha_i + x_i)} \prod_{i=1}^{k} p_i^{(\alpha_i + x_i - 1)}$$

$$P(p|x) \propto \frac{\prod_{i=1}^{k} p_i^{(\alpha_i + x_i - 1)}}{B(\alpha + x)}$$

where $B(\alpha + x)$ is the normalization constant for the Dirichlet distribution with updated parameters $\alpha_i + x_i$. The expectation of each p_i in the posterior distribution is given by:

$$E(p_i) = \frac{\alpha_i + x_i}{\sum_{i=1}^{k} \alpha_i + x_i}$$

As new information becomes available, we can continue updating our beliefs by using the previously computed posterior as the new prior, iteratively refining our inference about p_i. We encourage the reader to refer to the appendix for an example of how the dirichlet would be used as a conjugate prior of the multinomial in a real case.

Understanding the Dirichlet distribution as a conjugate prior for the Multinomial distribution is crucial for implementing Latent Dirichlet Allocation (LDA) In LDA, we model the distribution of topics over words in a set of documents, where each document is treated as a mixture of topics, and each topic as a distribution over words.

3.3. Markov Chain Monte Carlo (MCMC) methods

Markov Chain Monte Carlo (MCMC) methods are powerful techniques often used in Bayesian inference. While *traditional* Monte Carlo methods rely on generating independent samples, MCMC methods create a sequence of dependent samples. This dependency allows MCMC methods to be effective when working with complex probability distributions that are otherwise challenging to sample from directly.

To understand MCMC, it's essential to grasp the basics of the Monte Carlo method. Monte Carlo methods approximate characteristics of the probability distribution (such as the expected value of a random variable X) when they cannot be computed analytically Y. F. Zhang and Yu (2010).

The key difference in MCMC is that the generated samples $x_1, \ldots, x_t$ are correlated, forming a sequence governed by the properties of a Markov chain. This Markov chain dependency is detailed further in the appendix.

MCMC methods are especially useful for Bayesian analysis, as they allow for the approximation of posterior distributions when conjugate priors or analytical solutions are not available. By constructing a Markov chain that has the desired distribution as its equilibrium distribution, MCMC methods, such as the Gibbs sampler and Metropolis-Hastings algorithm, provide a way to sample from complex posterior distributions.

3.4. Gibbs Sampling

Gibbs sampling is a specific MCMC algorithm used to sample from a multivariate probability distribution when direct sampling from the joint distribution is difficult. Instead, it samples from each variable's conditional distribution sequentially, making it a practical tool for Bayesian inference in high-dimensional models (Gelfand 2000).

1. **Why is it a Markov Chain?** In Gibbs sampling, each iteration generates a new sample based on the current sample. The conditional distribution of each variable, given the values of the other variables, is used to sample, adhering to the Markov property: the next state depends only on the current state, not on past states. This creates a Markov chain where each sample is linked to the previous one.

2. **Why is it a Monte Carlo Simulation?** Gibbs sampling is a Monte Carlo technique because it relies on generating random samples to approximate characteristics of the target distribution. Monte Carlo methods typically estimate integrals, probabilities, and other quantities using random samples. Gibbs sampling approximates the joint distribution of variables by generating samples from the conditional distributions of each variable in turn.

Consider a joint density function $f(x, y_1, \ldots, y_p)$ where the objective is to determine characteristics of the marginal density, such as the mean or variance. A straightforward approach would involve computing $f(x)$ directly and using it to derive the desired properties:

$$f(x) = \int \ldots \int f(x, y_1, \ldots, y_p) dy_1 \ldots dy_p$$

However, in many scenarios, these integrals are either intractable and exceedingly difficult to to evaluate analytically or numerically. In such cases, Gibbs sampling provides an efficient alternative for estimating $f(x)$. Rather than explicitly

computing or approximating $f(x)$ directly, Gibbs sampling enables the generation of a sample $X_1, \ldots, X_M \sim f(x)$ without requiring the explicit functional form of $f(x)$ (Carpenter 2010). From this simulated sample, the mean, variance, or any other characteristic of $f(x)$ can be estimated.

Notably, the results of calculations, even when based on simulations, pertain to population-level quantities. For example, the mean of $f(x)$, can be approximated using the sample mean $\frac{1}{m}\sum_{i=1}^{m} X_i$ leveraging the following theoretical result:

$$\lim_{m\to\infty} \frac{1}{m}\sum_{i=1}^{m} X_i = \int_{-\infty}^{\infty} x f(x) dx = E[X]$$

By ensuring that the sample size m is sufficiently large, it becomes possible to approximate any parameter of $f(x)$ —and even the density itself— with a high degree of accuracy.

To understand how Gibbs sampling works, let's begin with a pair of random variables (X, Y). Gibbs sampling generates a sample from $f(x)$ using the conditional probabilities $f(x|y)$ and $f(y|x)$, which are often readily available in statistical models. This is achieved by constructing a 'Gibbs sequence' of random variables:

$$Y'_0, X'_0, Y'_1, X'_1, Y'_2, X'_2, \ldots, Y'_k, X'_k.$$

The process starts by assigning an arbitrary initial value $Y'_0 = y'_0$ and the subsequent values are computed iteratively as follows:

$$X'_j \sim f(x|Y'_j = y'_j)$$

$$Y'_{j+1} \sim f(y|X'_j = x'_j)$$

By averaging the conditional densities along the Gibbs sequence, an estimate of the density can be obtained. Specifically, the mean of the conditional densities $f(x|Y'_k = y'_k)$ provides an approximation to $f(x)$:

$$\hat{f(x)} = \frac{1}{m}\sum_{k=1}^{m} f(x|y'_i)$$

where $y'_1, \ldots, y'_m$ is the sequence of values generated for Y during Gibbs sampling. This method relies on the theoretical property:

$$E[f(x|Y)] = \int f(x|y) f(y) dy = f(x)$$

For scenarios involving more than two variables, Gibbs sampling extends naturally by iterating over all conditional distributions. For example, with three variables X, Y, Z the j-th iteration proceeds as follows:

$$X'_j \sim f(x|Y'_j = y'_j, Z'_j = z'_j)$$

$$Y'_{j+1} \sim f(y|X'_j = x'_j, Z'_j = z'_j)$$

$$Z'_{j+1} \sim f(z|X'_j = x'_j, Z'_{j+1} = z'_{j+1})$$

3.4.1. Application to LDA

Gibbs sampling is crucial for implementing Latent Dirichlet Allocation (LDA). In this context, the algorithm requires calculating the probabilities of latent variables, which represent the topics associated with documents and the words within those topics. Due to the high dimensionality and complexity of the model, directly computing these probabilities is often infeasible.

Gibbs sampling simplifies this task by iteratively drawing samples from the conditional distributions of the latent variables. This allows the posterior distribution of the topic assignments for each word to be estimated effectively (Darling 2011). Through these iterative updates, the sampler converges to the true underlying topic structure, enabling accurate inference in LDA. Further details about the implementation of the Gibbs sampler in LDA will be discussed in Section 5.

4. Text Preprocessing and word representation for LDA

In the previous sections, we introduced key concepts in probability theory and the Bayesian framework, which form the foundation of Latent Dirichlet Allocation (LDA). In this section, we focus on text preprocessing and word representation—critical steps for applying LDA to a corpus of documents. We introduce fundamental concepts from natural language processing (NLP), such as the Bag of Words (BoW) model, the principle of exchangeability, and preprocessing techniques like stemming and lemmatization.

Efficiently representing text data is crucial for the successful implementation of LDA, as these representations directly influence the quality of the inferred topics. We will also discuss the use of unigrams and bigrams and their implications for topic modeling.

4.1. Bag of Words and Exchangeability

A foundational step in implementing LDA within is deciding how to represent words within documents. One of the most commonly used techniques is the Bag of Words (BoW) model, which assumes that the order of words in a document is irrelevant and only captures the frequency of each word (Qader et al. 2019).

This assumption aligns with the principle of exchangeability, which states that the joint probability distribution of a set of random variables remains unchanged under permutations of their order. In the context of LDA, exchangeability allows us to prioritize word counts over word order, simplifying the computational model while preserving the essential topic structure of documents.

In the BoW model, documents are represented as vectors, where each element corresponds to the frequency of a specific word. LDA operates on these vectors to infer topics, treating topics as distributions over words (Aman Kedia 2020). Each document is modeled as a mixture of topics, and each topic as a multinomial distribution over words. By ignoring word order, the BoW representation reduces the complexity of modeling word-topic distributions while maintaining focus on the core structure of the data.

4.2. Unigrams and Bigrams

In LDA, words are typically modeled as unigrams, where each word is treated as an individual token. However, in some cases, the use of bigrams (pairs of adjacent words) can enrich the context and capture multi-word expressions that convey distinct meanings (e.g., "machine learning"). The inclusion of bigrams can enhance the quality of the topics inferred by LDA, as certain concepts are often expressed as fixed word pairs rather than individual words (Kaur et al. 2017).

For instance, in a technology-related corpus, the bigram neural network carries a very different meaning compared to the individual words neural and network. By incorporating bigrams into the model, LDA can better recognize such fixed expressions, improving the interpretability and coherence of the inferred topics.

However, incorporating bigrams comes with trade-offs. It increases the dimensionality of the word space, which may lead to higher computational costs. Deciding whether to include bigrams depends on the specific application and the nature of the corpus (Braga et al. 2003).

4.3. Stemming and Lemmatization

To optimize word representation in LDA, preprocessing techniques like stemming and lemmatization are often applied.

- Stemming reduces words to their root form (e.g., running becomes run).
- Lemmatization converts words to their dictionary form (e.g., better becomes good).

Both techniques group different forms of the same word, reducing the dimensionality of the word space and improving the LDA's performance. This step is essential, as it allows the model to focus on core meanings of words rather than being distracted by superficial variations. By applying stemming or lemmatization before creating the BoW representa- tion, we ensure that words with similar meanings are treated as a single entity. This is particularly important for large corpora,

as it reduces noise and enhances the coherence of the inferred topics (Balakrishnan and Lloyd-Yemoh 2014).

4.4. Application in LDA

Word representation plays a pivotal role in the implementation of LDA, directly influencing how topics are extracted from the data. The BoW model, combined with preprocessing techniques like stemming and lemmatization, simplifies text data and makes it more amenable to topic modeling. Additionally, the decision to use unigrams or bigrams depends on the desired granularity of topics and the context of the corpus.

By employing these word representation techniques, LDA can uncover the latent structure of text, revealing patterns and relationships between words and topics that might otherwise remain hidden. This structured approach ensures that LDA can effectively model complex corpora, facilitating deeper insights into the underlying themes within the data.

5. Latent Dirichlet Allocation

In previous sections, we covered the fundamental concepts of probability theory, introduced the Bayesian framework essential for understanding LDA, and discussed key concepts of natural language processing (NLP). Now, we turn our focus to LDA itself—a generative model used to uncover hidden topics within a collection of documents.

5.1. LDA as a generative model

It is crucial to understand that LDA can be approached from two primary perspectives: as a generative model and through inference techniques. LDA provides a probabilistic framework describing how a set of documents could be generated given underlying topics. The core assumption in LDA is that each document is a mixture of topics, and each topic is represented as a distribution over words.

5.1.1. The model and Terminology

This section explains the generative process underlying the LDA model. Readers are encouraged to thoroughly review Table 1, which summarizes the variable names and their meanings in LDA, before proceeding. The general structure of the LDA model is depicted in Figure 4.

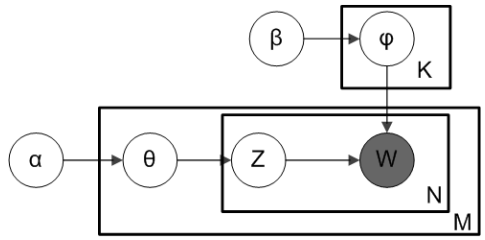

FIGURE 4: LDA Model, Source: Blei et al. (2003)

Variable	Type	Meaning
K	integer	Number of topics (e.g., 50)
V	integer	Number of words in vocabulary (e.g., 50,000 or 1,000,000)
M	integer	Number of documents
$N_{d=1,\ldots M}$	integer	Number of words in document d
N	integer	Total number of words; $N = \sum_{d=1}^{M} N_d$
$\alpha_{k=1\ldots K}$	positive real	Prior weight of topic k; often $\alpha < 1$, e.g., 0.1
α	K-dimensional vector	Collection of all α_k values
$\beta_{w=1\ldots V}$	positive real	Prior weight of word w; usually $\beta \ll 1$, e.g., 0.001
β	V-dimensional vector	Collection of all β_w values
$\phi_{k=1\ldots K,w=1\ldots V}$	probability	Probability of word w in topic k
$\phi_{k=1\ldots K}$	V-dimensional vector	Distribution of words in topic k
$\theta_{d=1\ldots M,k=1\ldots K}$	probability	Probability of topic k in document d
$\theta_{d=1\ldots M}$	K-dimensional vector	Distribution of topics in document d
$z_{d=1\ldots M,n=1\ldots N}$	integer	Topic identity of word n in document d
Z	N-dimensional vector	Topic identity of all words in all documents
$w_{d=1\ldots M,n=1\ldots N}$	integer	Identity of word n in document d
W	N-dimensional vector	Identity of all words in all documents

TABLE 1: Variables and their meaning in Latent Dirichlet Allocation (LDA). Source: Authors

LDA assumes the following generative process for each document in a corpus:

1. $N \sim \text{Poisson}(\xi)$: The number of words N in a document is drawn from a Poisson distribution with parameter ξ.
2. $\theta \sim \text{Dirichlet}(\alpha)$: The topic distribution θ for a document is drawn from a Dirichlet distribution with parameter α.
3. For each topic $k \in \{1, \ldots, K\}$:
   - $\phi_k \sim \text{Dirichlet}(\beta)$: The word distribution for topic k is drawn from a Dirichlet distribution with parameter β.

4. For each of the N words $w_{d,w}$ in document d:
   - Draw $z_{d,n} \sim \text{Multinomial}(\theta)$: The topic of word $w_{d,n}$ is drawn from a multinomial distribution with parameter θ.
   - Draw $w_{d,n} \sim \text{Multinomial}(\phi_{k,w})$: The word $w_{d,n}$ is drawn from the topic-word distribution $\phi_{k,w}$ given the topic $z_{d,n}$.

This process defines how documents could be generated from topics and distributions, offering a probabilistic model for the corpus. Further details are explained in this section.

5.1.2. Model Explanation and Visualizations

The joint distribution of topic mixtures θ, topic assignments z, and words w given the parameters α and β, is defined as:

$$p(\phi, \theta, \mathbf{z}, \mathbf{w} \mid \alpha, \beta) = \prod_{k=1}^{K} p(\phi_k \mid \beta) \prod_{d=1}^{M} p(\theta_d \mid \alpha) \prod_{n=1}^{N} p(z_{d,n} \mid \theta_d) p(w_{d,n} \mid z_{d,n}, \phi_{k,w})$$

For further explanation, the reader is referred to the Appendix.

1. **Topic Distribution in Documents:** Assuming a vocabulary with three words and a corpus containing three topics, the topic distribution $p(\theta_d \mid \alpha)$ of θ_d for a document d, drawn from a Dirichlet distribution, can be represented as in Figure 5.

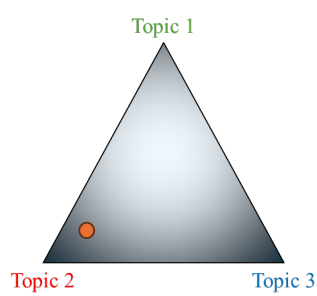

FIGURE 5: Representation of the topic distribution of document d. Source: Authors

Each point inside the triangle represents the probabilities of Topics 1, 2, and 3 for d. For instance, $\theta_d = (0.2, 0.7, 0.1)$. Note that the parameter α influences this distribution within the simplex given the properties of the Dirichlet distribution used as a prior. Lower values of α promote sparsity, making the model more interpretable by ensuring most documents are dominated by a few topics.

Given this topic distribution, $p(z_{d,n} \mid \theta_d)$ can be visually represented as a bag with a certain number of items, let's assume 10 with a proportion of items $\theta_d = (0.2, 0.7, 0.1)$.

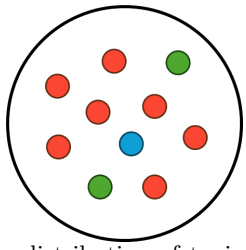

FIGURE 6: Representation of the distribution of topics in document d given the topic distribution of document d. Topics 1,2, and 3 are represented as green, red and blue circles respectively. Source: Authors

2. **Word Distributi on in Topics:** Similarly, word distributions for each topic ϕ_k are drawn from a Dirichlet distribution, denoted as $p(\phi_k \mid \beta)$ which represents the probability distribution of words within each topic. Given that our vocabulary consists of only three words, the word distribution for each topic is visualized within a triangle, which corresponds to a 2-simplex. However, if an additional word is introduced to the vocabulary, the word distributions would then be represented within a tetrahedron, as the dimensionality of the simplex increases. Refer to Figure 7 for a visual representation.

   To illustrate, let us assume the word distributions for Topics 1, 2, and 3 are as follows:

$$\phi_1 = (0.2, 0.6, 0.2), \quad \phi_2 = (0.9, 0.1, 0), \quad \phi_3 = (0.1, 0.2, 0.7)$$

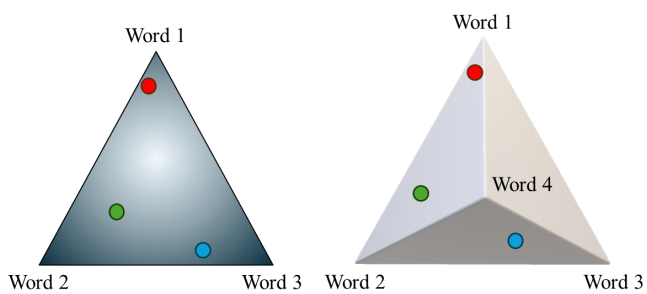

FIGURE 7: Representation of the distribution of words per topic in a vocabulary with 3 and 4 words. Source: Authors

Note that the parameter β influences this distribution within the simplex given the properties of the Dirichlet distribution used as a prior. Lower β values also ensure sparsity, making most topics dominated by a small

subset of words, thereby enhancing interpretability. With the ϕ_k vector of probabilities obtained, it is possible to represent these probabilities like a bag where we will sample in the future the word n of a document d given a determinate topic.

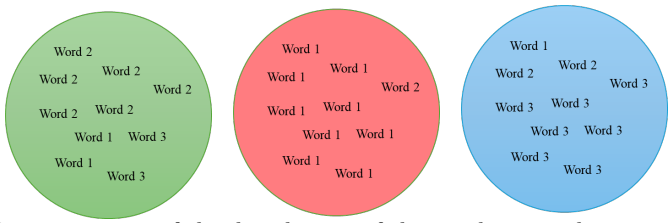

FIGURE 8: Representation of the distribution of the words given the topic 1, 2 or 3 of a document d. Source: Authors

3. **Summary:** In summary, once we have the distribution of topics per document (θ_d, as shown in Figure 5), we draw a topic ($z_{d,n}$) for the first word in document d based on the topic distribution θ_d (Figure 6). This step essentially assigns a topic to the word we are generating.

   Simultaneously, once we have the distribution of words per topic (ϕ_k , as shown in Figure 7), we draw the first word of document d based on the assigned topic ($z_{d,n}$) and the word distribution for that topic (ϕ_k). This process can be visualized as selecting a "bag" from Figure 8 corresponding to the assigned topic and then drawing a word from that bag. We repeat this process iteratively for each word until the document is complete, and this process is then repeated for all documents in the corpus.

   While the generative process describes how documents are constructed from topics, real-world application typically require solving inverse problem: documents, we aim to infer the underlying topics and their associated word distributions. In the next section, we will explore the techniques and algorithms used to perform inference in LDA, enabling the discovery of latent topics from unstructured text data.

5.2. LDA Inference

Latent Dirichlet Allocation (LDA) inference is critical for identifying the latent structure in a collection of documents when the topic distributions per document and word distributions per topic are unknown. The objective of LDA inference is to estimate these distributions using only the observed corpus. This process involves uncovering the hidden variables (latent variables), such as the topic proportions for each document and the word distributions associated with each topic.

5.2.1. Common Methods for LDA Inference

Several widely adopted methods facilitate LDA inference:

- *Collapsed Gibbs Sampling*: A Markov Chain Monte Carlo (MCMC) method that iteratively assigns topics to words and updates these assignments based on posterior distributions. This method is scalable and relatively straightforward to implement for large datasets.
- *Variational Inference*: An optimization-based approach that approximates the true posterior distribution with a simpler one. While faster than Gibbs Sampling, it may be less precise.

The following sections focus on implementing LDA using Collapsed Gibbs Sampling.

5.2.2. The Number of Topics

Choosing an appropriate number of topics (k) is essential, as it directly affects the interpretability and quality of the LDA results (Gan and Qi 2021). In LDA, k is treated as a hyper-parameter that must be determined prior to model training. Several strategies exist for estimating this parameter:

- **Perplexity Minimization**:

  Perplexity measures how well the model predicts unseen data. Lower perplexity indicates better generalization performance, suggesting that the chosen number of topics is appropriate (Popel and Mareček 2010). The formula for perplexity on a test dataset (D_{test}) is:

  $$\text{Perplexity}(D_{\text{test}}) = \exp\left(-\frac{\sum_{d\in D_{\text{test}}} \log p(w_d)}{\sum_{d\in D_{\text{test}}} N_d}\right)$$

  where N_d denotes the number of words in document d.

- **Coherence Score**:

  This metric evaluates the semantic similarity of high-probability words within a topic and often aligns more closely with human judgment compared to perplexity, making it a preferred choice for estimating the number of topics (Syed and Spruit 2017).

- **Elbow Method**:

  This technique plots a performance metric (e.g., perplexity or coherence) against k. The "elbow" point, where improvement slows significantly, indicates an optimal balance between model complexity and performance (Khalid and Wade 2020).

- **Cross-Validation**:

  By splitting the dataset into training and validation sets, one can determine the k that optimizes performance on the validation set. That's to say, one can train LDA models with different numbers of topics and select the model that

performs best on the validation set based on a chosen metric (Yalamanchili et al. 2017).

- **Hierarchical Dirichlet Process (HDP)**:

  It is a non-parametric Bayesian extension of LDA that automatically determines the number of topics based on the data. It enables the number of topics (k) to grow dynamically with the corpus size, adapting to the data's complexity without the need for a predefined k (Teh et al. 2004).

Other approaches, such as Expert Judgment and Interpretability (where domain experts manually evaluate the quality of topics), and the Bayesian Information Criterion (BIC, which balances model likelihood with complexity), also assist in determining k, balancing interpretability and statistical rigor.

5.2.3. Implementation of LDA collapsed Gibbs sampler

Once k is selected, the goal is to estimate:

- θ_d: Topic distributions per document.
- ϕ_k: Word distributions per topic.
- z_i: Topic assignments for each word.

5.2.4. Collapsed Gibbs Sampling Process

While conditional distributions and, consequently, an LDA Gibbs Sampling algorithm can be derived for each of these latent variables, it is noteworthy that both θ_d and ϕ_k can be computed solely from the topic index assignments z_i (i.e., z is a sufficient statistic for these distributions). This allows for a more streamlined approach by integrating out the multinomial parameters and focusing exclusively on sampling z_i. This approach is known as the collapsed Gibbs sampler (Darling 2011). In the collapsed Gibbs sampler for LDA, the central task is to calculate the probability of assigning a topic z to a word $w_{d,n}$, conditioned on all other topic assignments for the remaining words. Formally, the goal is to estimate the following posterior probability up to a normalization constant:

$$p(z_i|z_{-i}, \alpha, \beta, \mathbf{w})$$

where z_{-i} represents all topic assignments except for z_I. According to the rules of conditional probability, we can express the conditional distribution as follows:

$$p(z_i|z_{-i}, \alpha, \beta, \mathbf{w}) = \frac{p(z_i, z_{-i}, \mathbf{w}|\alpha, \beta)}{p(z_{-i}, \mathbf{w}|\alpha, \beta)} \propto p(z_i, z_{-i}, \mathbf{w}|\alpha, \beta) = p(\mathbf{z}, \mathbf{w}|\alpha, \beta)$$

The Gibbs sampling equation for LDA can then be derived by applying the chain rule. For simplicity, we omit the hyperparameters α and β in the notation.

The detailed derivation and intermediate mathematical steps can be found in Darling's article (2011), as cited in the bibliography.

$$p(z_i|z^{(-i)}, \mathbf{w}) = \frac{p(\mathbf{w}, \mathbf{z})}{p(\mathbf{w}, z^{(-i)})} = \frac{p(\mathbf{z})}{p(z^{(-i)})} \frac{p(\mathbf{w}|\mathbf{z})}{p(\mathbf{w}^{(-i)}|z^{(-i)})} p(w_i)$$

$$\propto \prod_d \frac{B(n_{d,\cdot} + \alpha)}{B(n_{d,\cdot}^{(-i)} + \alpha)} \prod_k \frac{B(n_{k,\cdot} + \beta)}{B(n_{k,\cdot}^{(-i)} + \beta)}$$

$$\propto \prod_d \frac{\Gamma(n_{d,k} + \alpha_k)\Gamma(\sum_{k=1}^K n_{d,k} + \alpha)}{\Gamma(n_{d,k}^{(-i)} + \alpha_k)\Gamma(\sum_{k=1}^K n_{d,k}^{(-i)} + \alpha)} \prod_k \frac{\Gamma(n_{k,w} + \beta_w)\Gamma(\sum_{w=1}^W n_{k,w} + \beta)}{\Gamma(n_{k,w}^{(-i)} + \beta_w)\Gamma(\sum_{w=1}^W n_{k,w}^{(-i)} + \beta)}$$

$$\propto \frac{(n_{d,k}^{(-i)} + \alpha_k)(n_{k,w}^{(-i)} + \beta_w)}{\sum_{w'} n_{k,w'}^{(-i)} + \beta_{w'}} \tag{1}$$

The collapsed Gibbs sampler is a popular inference method for Latent Dirichlet Allocation (LDA) because it allows efficient estimation of the model's latent variables by integrating out (collapsing) the document-topic and topic-word distributions. Here's a summary of the implementation process:

1. **Initialize Topic Assignments**
2. **Iterate through Words**: For each word in the corpus, perform the following steps:
   a) **Remove Current Topic Assignment**: Temporarily remove the current topic of the word (this is often referred to as "leaving-one-out").
   b) **Calculate Conditional Probabilities**: Estimate the probability of assigning each topic to the word using equation 1. This probability is computed based on two factors:
      - **Document-Topic Proportion**: The probability that the document the word belongs to will generate a word from a given topic.
      - **Topic-Word Proportion**: The probability that a given topic will generate the specific word type.
   c) **Sample a New Topic**: Draw a new topic for the word based on the calculated conditional probabilities. Update the topic count for both the document and the topic.
3. **Repeat for Multiple Iterations**: Repeat this process for all words in the corpus over many iterations, allowing the sampler to converge to a stable distribution.
4. **Estimate Parameters**: After convergence, use the collected topic assignments to estimate the document-topic distributions θ and topic-word distributions ϕ as the final outputs of the model.

This approach is called "collapsed" Gibbs sampling because it marginalizes out θ and ϕ and focuses solely on the topic assignments for each word, making the process computationally more efficient.

5.2.5. Final Results and Applications

Once the model stabilizes, LDA provides:

- Topic Distributions per Document (θ_d): Useful for clustering documents based on thematic composition.
- Word Distributions per Topic (ϕ_k): Enables identification of the most relevant words for each topic.

These results can be utilized in various ways:

- **Document Clustering:** The θ distributions allow for clustering documents based on their topic compositions. This can be useful in organizing large datasets into coherent groups, See example in bibliography (Onan et al. 2017).
- **Topic Visualization:** The ϕ distributions provide insights into the most relevant words for each topic, which can be visualized using word clouds or bar charts to enhance interpretability, See example in bibliography (Yang et al. 2016).
- **Content Recommendation:** By analyzing the topics present in user-generated content, systems can recommend similar articles, enhancing user engagement, See example in bibliography Bagul and Barve (2021).
- **Trend Analysis:** Over time, the evolution of topics can be studied to understand trends within a specific domain, which is valuable for researchers and industry professionals alike, See example in bibliography (Zou et al. 2024).
- **Social Media Analysis:** Companies have employed LDA to analyze social media posts and extract prevalent topics, aiding in marketing strategies and public sentiment analysis, See example in bibliography (Wang et al. 2022).
- **Scientific Research:** LDA has been utilized in various studies to discover emerging research trends and analyze the evolution of scientific topics across publications, See example in bibliography (Wu et al. 2014).
- **News Aggregation:** News organizations apply LDA to categorize articles into topics, helping users navigate vast amounts of information efficiently, See example in bibliography (Hamborg et al. 2017).

Final Remarks

This document is designed to serve as a comprehensive guide for individuals seeking to start from scratch with Latent Dirichlet Allocation (LDA). It provides a structured learning path, featuring clear explanations, practical examples, and mathematical demonstrations, ensuring that readers establish a strong foundation before diving into the LDA model itself.

Throughout the chapters, we have covered the essential concepts and methodologies related to LDA, equipping readers with the necessary tools to implement and analyze topic models effectively from the ground up. As the field of data science continues to evolve, topic modeling remains a crucial technique. By utilizing LDA, researchers and analysts can uncover hidden patterns within textual data, leading to valuable insights and informed decision-making. We hope this guide inspires readers to further explore advanced topics in machine learning and natural language processing, cultivating a deeper understanding of the potential and applications of topic modeling.

For any suggestions, questions, or comments, please feel free to contact us at carlosdl@uninorte.edu.co or hllinas@uninorte.edu.co.

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Appendix A. Properties of the Poisson Distribution

- $E(X) = \lambda$

Proof: Let $X \sim P(\lambda)$, the expectation of X is given by:

$$E(X) = \sum_{x=0}^{\infty} x f_X(x) = \sum_{x=0}^{\infty} \frac{xe^{-\lambda}}{x!}\lambda^x$$

Evaluating the first term gives:

$$E(X) = \frac{0e^{-\lambda}}{0!}\lambda^0 + \sum_{x=1}^{\infty} \frac{xe^{-\lambda}}{x(x-1)!}\lambda^x$$

$$E(X) = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!}$$

By making the variable change $s = x - 1$, we have:

$$E(X) = e^{-\lambda} \sum_{s=0}^{\infty} \frac{\lambda^{s+1}}{s!} = \lambda e^{-\lambda} \sum_{s=0}^{\infty} \frac{\lambda^s}{s!}$$

Given that $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$;

$$E(X) = \lambda e^{-\lambda} \sum_{s=0}^{\infty} \frac{\lambda^s}{s!} = \lambda e^{-\lambda} e^{\lambda} = \lambda \quad \blacksquare$$

- $m_X(t) = e^{\lambda(e^t - 1)}$

Proof: Using the moment method, we have that:

$$m_X(t) = E(e^{Xt}) = \sum_{x=0}^{\infty} e^{xt}.f_X(x) = \sum_{x=0}^{\infty} e^{xt}.\frac{\lambda^x e^{-\lambda}}{x!}$$

$$m_X(t) = e^{-\lambda} \sum_{x=0}^{\infty} \frac{e^{xt}\lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!}$$

$$m_X(t) = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)} \quad \blacksquare$$

- $Var(X) = \lambda$

Proof: Given the moment generating function derived earlier, we know from properties that $E[X] = m'_X(0)$ and $E[X^2] = m''_X(0)$. Therefore:

$$m'(t=0) = [e^{\lambda(e^t-1)}\lambda e^t]\Big|_{t=0} = e^{\lambda(e^0-1)}\lambda e^0 = \lambda$$

Now, calculating the second derivative:

$$m''(t) = e^{\lambda(e^t-1)}\lambda^2 e^{2t} + e^{\lambda(e^t-1)}\lambda e^t$$

$$m''(0) = e^{\lambda(e^0-1)}\lambda^2 e^0 + e^{\lambda(e^0-1)}\lambda e^0 = \lambda^2 + \lambda$$

Using the property $Var(X) = E[X^2] - E[X]^2$:

$$Var(X) = m''(0) - m'(0)^2$$

$$Var(X) = \lambda^2 + \lambda - \lambda^2 = \lambda \quad \blacksquare$$

Appendix B. Parameter estimation of Poisson Distribution

Let $X = X_1, X_2, ...X_n$

$$f_X(x_i) = \begin{cases} \frac{e^{-\lambda}}{x!}\lambda^x & , x_i = 0,1,2,..,n \\ 0 & , \text{Otherwise} \end{cases}$$

Assembling the joint probability function and setting the x_i.

$$L(\lambda) = f_X(x_1, x_2, ..., x_n) = \prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

$$\mathcal{L}(\lambda) = ln(\prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}) = \sum_{i=1}^{n} ln(\frac{\lambda^{x_i} e^{-\lambda}}{x_i!})$$

$$\mathcal{L}(\lambda) = \sum_{i=1}^{n} [ln(\lambda^{x_i}) + ln(e^{-\lambda}) - ln(x_i!)]$$

$$\mathcal{L}(\lambda) = \sum_{i=1}^{n} [x_i \cdot ln(\lambda) - \lambda \cdot ln(e) - ln(x_i!)]$$

$$\mathcal{L}(\lambda) = ln(\lambda)\sum_{i=1}^{n} x_i - \sum_{i=1}^{n}\lambda - \sum_{i=1}^{n} ln(x_i!)$$

$$\mathcal{L}(\lambda) = ln(\lambda)\sum_{i=1}^{n} x_i - n\lambda - \sum_{i=1}^{n} ln(x_i!)$$

Deriving and equalling 0 we have:

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \frac{\sum_{i=1}^{n} x_i}{\hat{\lambda}} - n = 0$$

$$\hat{\lambda} = \frac{\sum_{i=1}^{n} x_i}{n} \quad \blacksquare$$

Deriving a second time, it is found that this estimator maximises the likelihood since it is found that:

$$\frac{\partial^2 \mathcal{L}}{\partial^2 \lambda} = -\frac{\sum_{i=1}^{n} x_i}{\lambda^2} > 0$$

It is also found that this estimator is unbiased given that $E(\hat{\lambda}) = \lambda$

$$E(\hat{\lambda}) = \frac{1}{n}E(\sum_{i=1}^{n} x_i) = \frac{1}{n}\sum_{i=1}^{n} E(x_i)$$

$$E(\hat{\lambda}) = \frac{n\lambda}{n} = \lambda \quad \blacksquare$$

Appendix C. Properties of the Beta Distribution

Theorem. For all $n > -\alpha$:

$$E(X^n) = \frac{\Gamma(\alpha+n)\Gamma(\alpha+\beta)}{\Gamma(\alpha+\beta+n)\Gamma(\alpha)} = \frac{B(n+\alpha,\beta)}{B(\alpha,\beta)} = \prod_{r=0}^{n-1}\frac{\alpha+r}{\alpha+\beta+r}$$

Proof: By definition of expectation, we have

$$E(X^n) = \int_{-\infty}^{\infty} x^n f(x)\cdot dx$$

$$E(X^n) = \frac{1}{B(\alpha,\beta)}\int_0^1 x^n \cdot x^{\alpha-1}(1-x)^{\beta-1}\cdot dx$$

$$E(X^n) = \frac{1}{B(\alpha,\beta)}\int_0^1 x^{n+\alpha-1}(1-x)^{\beta-1}\cdot dx = \frac{B(n+\alpha,\beta)}{B(\alpha,\beta)}$$

$$E(X^n) = \frac{\Gamma(\alpha+n)\Gamma(\beta)/\Gamma(\alpha+\beta+n)}{\Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)}$$

$$E(X^n) = \frac{\Gamma(\alpha+n)\cdot\Gamma(\alpha+\beta)}{\Gamma(\alpha+\beta+n)\cdot\Gamma(\alpha)} \quad \blacksquare$$

- $E(X) = \frac{\alpha}{\alpha+\beta}$

Proof: Let $X \sim B(\alpha,\beta)$ be the expectation of X^n with $n=1$, which is given by:

$$E(X) = \frac{\Gamma(\alpha+1)\cdot\Gamma(\alpha+\beta)}{\Gamma(\alpha+\beta+1)\cdot\Gamma(\alpha)}$$

By the property of the Gamma function, $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$, then:

$$E(X) = \frac{\alpha\cdot\Gamma(\alpha)\cdot\Gamma(\alpha+\beta)}{(\alpha+\beta)\cdot\Gamma(\alpha+\beta)\cdot\Gamma(\alpha)} = \frac{\alpha}{\alpha+\beta} \quad \blacksquare$$

- $Var(X) = \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$

Proof: Let $X \sim B(\alpha,\beta)$ be the expectation of X^n with $n=2$, which is given by:

$$E(X^2) = \frac{\Gamma(\alpha+2)\cdot\Gamma(\alpha+\beta)}{\Gamma(\alpha+\beta+2)\cdot\Gamma(\alpha)}$$

By the property of the Gamma function, $\Gamma(\alpha+2) = (\alpha+1)\cdot\Gamma(\alpha+1) = (\alpha+1)\cdot\alpha\Gamma(\alpha)$, then:

$$E(X^2) = \frac{(\alpha+1)\cdot\alpha\Gamma(\alpha)\cdot\Gamma(\alpha+\beta)}{(\alpha+\beta+1)\cdot(\alpha+\beta)\cdot\Gamma(\alpha+\beta)\cdot\Gamma(\alpha)} = \frac{\alpha(\alpha+1)}{(\alpha+\beta+1)(\alpha+\beta)}$$

By definition of variance, $Var(X) = E(X^2) - E(X)^2$, then:

$$Var(X) = \frac{\alpha^2+\alpha}{(\alpha+\beta+1)(\alpha+\beta)} - \left(\frac{\alpha}{(\alpha+\beta)}\right)^2$$

$$Var(X) = \frac{(\alpha^2+\alpha)(\alpha+\beta)^2 - \alpha^2(\alpha+\beta+1)(\alpha+\beta)}{(\alpha+\beta+1)(\alpha+\beta)^3}$$

$$Var(X) = \frac{(\alpha+\beta)[(\alpha^2+\alpha)(\alpha+\beta) - \alpha^2(\alpha+\beta+1)]}{(\alpha+\beta+1)(\alpha+\beta)^3}$$

$$Var(X) = \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2} \quad \blacksquare$$

- **Moment Generating Function**

Proof:

$$M_X(t) = E(e^{tX}) = E\left(\sum_{x=0}^{\infty} \frac{(tx)^n}{n!}\right) = E\left(\sum_{x=0}^{\infty} \frac{t^n}{n!} \cdot x^n\right)$$

$$M_X(t) = \sum_{x=0}^{\infty} \frac{t^n}{n!} E(x^n) = \sum_{x=0}^{\infty} \left(\frac{t^n}{n!} \cdot \frac{B(n+\alpha,\beta)}{B(\alpha,\beta)}\right)$$

$$M_X(t) = \frac{t^0}{0!} \cdot \frac{B(\alpha,\beta)}{B(\alpha,\beta)} + \sum_{x=1}^{\infty} \left(\frac{t^n}{n!} \cdot \frac{B(n+\alpha,\beta)}{B(\alpha,\beta)}\right)$$

$$M_X(t) = 1 + \sum_{x=1}^{\infty} \left(\prod_{r=0}^{n-1} \frac{\alpha+r}{\alpha+\beta+r}\right) \frac{t^n}{n!} \quad \blacksquare$$

Appendix D. Properties of the Multinomial Distribution

- **Representation as a Sum of Multinoulli Vectors:** A random vector X that follows a multinomial distribution with parameters n and $p = p_1, p_2, \ldots, p_k$ can be expressed as:

  $$X = Y_1 + Y_2 + \ldots + Y_n$$

  where each Y_i is an independent variable with a multinoulli or categorical distribution, with $E(Y_i) = p_i$.

- $X_i \sim \text{Bin}(n, p_i)$

  The marginal distribution of the random variables X_i is binomial with parameters n and p_i, where $i = 1, 2, \ldots, k$, and they are not independent of each other.

- $E(X) = np$, where the vector p is defined as $p = \left\{p_1, p_2, \ldots, p_k\right\}^T$.

- $Var(X) = n\Sigma$, where Σ is a $K \times K$ matrix with values:

  $$\Sigma_{ij} = \begin{cases} np_i(1-p_i) & \text{, if } i = j \\ -np_ip_j & \text{, if } i \neq j \end{cases}$$

- **Moment Generating Function:** The moment generating function is given by:

  $$M_X(t) = (\sum_{i=1}^{k} p_i e^{t_i})^n$$

Proof: Since X can be expressed as the sum of n multinoulli vectors with parameters p, we have $X = Y_1 + Y_2 + \ldots + Y_n$:

$$M_X(t) = E(e^{t(Y_1+Y_2+\ldots+Y_n)}) = \prod_{i=1}^{n} E(e^t Y_i)$$

$$M_X(t) = \prod_{i=1}^{n} M_{Y_i}(t) = \left(\sum_{i=1}^{k} p_i e^{t_i}\right)^n \quad \blacksquare$$

Appendix E. Properties of the Dirichlet Distribution

- **Expectation:** The expectation of a Dirichlet random vector X is given by:

$$E[X] = \frac{1}{\sum_{k=1}^{K+1} \alpha_k} \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_K \end{bmatrix}$$

- **Covariance matrix:** The variance-covariance matrix of a Dirichlet random vector X is given by:

$$Cov[X_i, X_j] = \begin{cases} \frac{\alpha_i \sigma - \alpha_i^2}{\sigma^2(1+\sigma)} & \text{if } i = j \\ \\ -\frac{\alpha_i \alpha_j}{\sigma^2(1+\sigma)} & \text{if } i \neq j \end{cases}$$

- **Cross moments:** The cross moments of a Dirichlet random vector are:

$$E\left[\prod_{k=1}^{K} X_k^{\beta_k}\right] = \frac{\Gamma\left(\sum_{k=1}^{K+1} \alpha_k\right)}{\prod_{k=1}^{K+1} \Gamma(\alpha_k)} \cdot \frac{\Gamma(\alpha_{K+1}) \prod_{k=1}^{K} \Gamma(\alpha_k + \beta_k)}{\Gamma\left(\alpha_{K+1} + \sum_{k=1}^{K} (\alpha_k + \beta_k)\right)}$$

  where $\beta_1, \ldots, \beta_K$ are non-negative integers.

The proofs for these results can be found in the work by Marcos Taboga in 2021 Taboga (2021). See references.

Appendix F. Example of Binomial Model - Bayesian Approach

Given a sequence of an exchangeable binary variables $y = y_1, y_2, ...y_n$ where $i = \{1, 2, ..., n\}$ and all $y_i|\theta \sim Ber(\theta)$. In this model, per expert judgment, the θ parameter follows and uniform distribution $\theta \sim Unif(0, 1)$.

It is commonly used a DAG (Directed Acyclic Graph) to better understand the model since Bayesian models are naturally hierarchical because the parameter (θ in this case) is modeled and then the observations are modeled given θ

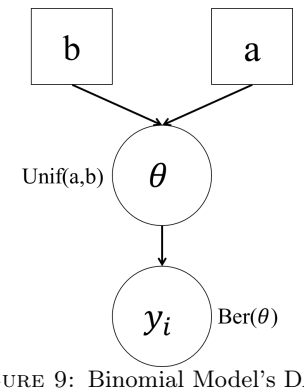

FIGURE 9: Binomial Model's DAG

Since θ is the parameter of the model, the prior can be expressed simply as $p(\theta)$. In order to build the posterior distribution $P(\theta|y)$ we must calculate the likelihood $P(y|\theta)$.

$$P(y|\theta) = P(y_1, y_2, ..., y_n|\theta) = \prod_{i=1}^{n} P(y_i|\theta) = \prod_{i=1}^{n} \theta^{y_i}(1-\theta)^{1-y_i}$$

$$P(y|\theta) = \theta^{\sum_{i=1}^{n} y_i}(1-\theta)^{n-\sum_{i=1}^{n} y_i}$$

where $s = \sum_{i=1}^{n} y_i$, then the likelihood can be written as follows:

$$P(y|\theta) = \theta^s(1-\theta)^{n-s}$$

Since we have already the prior and the likelihood, the posterior distribution can be written as follows:

$$P(\theta|y) = \frac{\theta^s(1-\theta)^{n-s} \cdot P(\theta)}{\int_\theta \theta^s(1-\theta)^{n-s} \cdot P(\theta)d\theta} \propto \theta^s(1-\theta)^{n-s} \cdot P(\theta)$$

since $\theta \in (0, 1)$ the expert can choose the prior depending of their judgment. Depending of this selection, posterior can drastically change as it depends on $p(\theta)$.

Appendix G. Gamma-Poisson Model

let's consider a state of knowledge with count variables $y = y_1, y_2, ..., y_n$ in an exchangeable sequence. Since all y_i are count variables, their probability distribution is Poisson with an unknown parameter λ which represents the unknown rate of occurrence for the events being modeled. Per our expert judgment we have the

belief that λ follows a Gamma distribution with parameters a and b. Therefore our model can be expressed as follows:

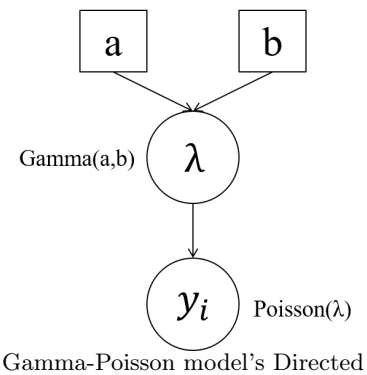

FIGURE 10: Gamma-Poisson model's Directed Acyclic Graph

$$\begin{cases} y_i|\lambda & \sim \ Poisson(\lambda), \quad i = \{1, 2, ..., n\} \\ \lambda & \sim \ Gamma(a, b) \end{cases}$$

Since $\lambda \sim Gamma(a, b)$, then $P(\lambda) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \cdot e^{-b\lambda}$. Therefore, the posterior can be expressed as follows:

$$P(\lambda|y) \propto P(y|\lambda) \cdot P(\lambda)$$

$$P(\lambda|y) \propto \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{y_i}}{y_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}$$

$$P(\lambda|y) \propto \lambda^s e^{-n\lambda} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}$$

where $s = \sum_{i=1}^{n} y_i$. Removing the constant term, the posterior can be written as:

$$P(\lambda|y) \propto \lambda^{(a+s)-1} e^{-\lambda(b+n)}$$

This is also known as the kernel of gamma distribution. In conclusion, under the Gamma-Poisson model, the posterior distribution of the parameter λ given the data y follows a Gamma distribution, specifically: $\lambda|y \sim Gamma(a+s, b+n)$ where a and b are the parameters of the prior Gamma distribution, s is the sum of the observed counts, and n is the number of observations.

The fact that the posterior distribution of λ given the data follows a Gamma distribution allows us to calculate its expected value easily. The expected value of a Gamma-distributed random variable with parameters a and b is given by $\mathbb{E}[\lambda] = a/b$. In our case, with the posterior distribution $\lambda \mid y \sim \text{Gamma}(a+s, b+n)$, the expected value can be computed as:

$$\mathbb{E}[\lambda \mid y] = \frac{a+s}{b+n}$$

This result provides a convenient way to estimate the rate parameter λ based on the observed data.

Appendix H. Simple example of Dirichlet as conjugate prior multinomial

We aim to model the purchase of products in a store. Random customers enter to the store, and they have certain probabilities of buying product 1, 2, 3, or any other product. Initially, we do not know the real proportions of what we will sell but we know that last time we sold 16 units of product 1, 12 of product 2, 8 of product 3 and 4 among other products. This means that we can set as priors $\alpha = [16, 12, 8, 4]$, corresponding to 40%, 30%, 20%, and 10%, respectively.

We collect data from 100 customers, and the sales are as follows:

- Product 1: 62
- Product 2: 21
- Product 3: 12
- Other products: 5

Quickly calculating the expectation formula, the probabilities derived from our posterior are:

$$E(p_1) = \frac{\alpha_i + x_i}{\sum_{i=1}^{k} \alpha_i + x_i} = \frac{62+16}{140} = 0.557$$

Similarly, for the other teams, we have:

- Product 1: 0.557
- Product 2: 0.236
- Product 3: 0.143
- Other products: 0.067

Had we computed the probabilities using maximum likelihood estimation, they would directly reflect the proportions observed in the data, without incorporating any prior knowledge. We notice that the resulting probabilities are still close to the data proportions but are slightly adjusted due to the prior. It is important to recognize that the influence of the priors depends on both the chosen α values and the size of old and new data.

In machine learning, this method is particularly valuable in cases where a new category appears in the test set. If the category was not present in the training set, it would be assigned a probability of zero, but using a prior can resolve this issue. Likewise, in NLP, the Dirichlet distribution is commonly employed as a prior.

Appendix I. Properties of Markov Chains

- **Lack of Memory:** A random sequence X_t is a Markov chain if and only if, given the current value X_t, future observations X_{t+n} are conditionally independent of past values X_{t-k} for any positive integers k and n.

  This property indicates that the process has no memory: the probability distribution of future values of the chain depends only on its current value X_t, regardless of how that value was reached (i.e., independent of the path taken by the chain).

$$P(X_{t+n} = x|X_t, X_{t-1}, \ldots, X_{t-k}) = P(X_{t+n} = x|X_t)$$

- **Irreducibility:**
  - ◇ In a Markov chain, a state e_j is said to be accessible from another state e_i if the probability of moving from state e_i to e_j at some future time is greater than zero.
  - ◇ A state e_i is communicated with another state e_j if e_j is accessible from e_i and e_i is accessible from e_j.
  - ◇ A Markov chain is said to be irreducible if all states communicate with each other.

- **Absorbing States:**
  - ◇ A state e_j is said to be absorbing if it is impossible to leave that state, i.e.:
$$P(X_k = e_j|X_{k-1} = e_j) = 1$$
  - ◇ A Markov chain is absorbing if it has any absorbing states.
  - ◇ If a Markov chain is irreducible and absorbing, then the probability of entering any of the absorbing states approaches one as the number of steps approaches infinity.

- **Regularity:**
  - ◇ A Markov chain is said to be regular if some power of the transition matrix has all its elements positive (no zeros).
  - ◇ If a chain is regular, then it is irreducible and non-absorbing.

Appendix J. LDA generative model

Given the Graphical model representation of LDA. The boxes are "plates" representing replicates. The outer plate represents documents, while the inner plate represents the repeated choice of topics and words within a document.

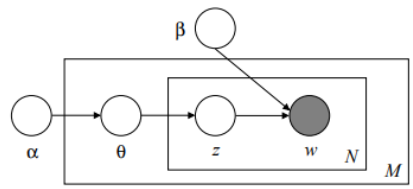

FIGURE 11: Graphical model representation of LDA

Given the parameters α and β, the joint distribution of a topic mixture θ, a set of N topics z, and a set of N words w is given by:

$$p(\theta, \mathbf{z}, \mathbf{w} \mid \alpha, \beta) = p(\theta \mid \alpha) \prod_{n=1}^{N} p(z_n \mid \theta) p(w_n \mid z_n, \beta)$$

Proof

$$p(\theta, \mathbf{z}, \mathbf{w} \mid \alpha, \beta) = \frac{p(\theta, \mathbf{z}, \mathbf{w}, \alpha, \beta)}{p(\alpha, \beta)}$$

Since $p(\theta \mid \beta, \alpha) = \frac{p(\theta,\alpha,\beta)}{p(\alpha,\beta)}$, we replace $p(\alpha, \beta)$.

Similarly, since θ does not depend on β, we have that $p(\theta \mid \beta, \alpha) = p(\theta \mid \alpha)$.

$$= \frac{p(\theta, \mathbf{z}, \mathbf{w}, \beta, \alpha) p(\theta \mid \alpha)}{p(\theta, \alpha, \beta)}$$

We multiply and divide by $p(\mathbf{z}, \theta, \alpha, \beta)$.

$$= \frac{p(\theta, \mathbf{z}, \mathbf{w}, \beta, \alpha)}{p(\theta, \alpha, \beta)} \cdot \frac{p(\mathbf{z}, \theta, \alpha, \beta)}{p(\mathbf{z}, \theta, \alpha, \beta)} \cdot p(\theta \mid \alpha)$$

$$= \frac{p(\theta, \mathbf{z}, \mathbf{w}, \beta, \alpha)}{p(\mathbf{z}, \theta, \alpha, \beta)} \cdot \frac{p(\mathbf{z}, \theta, \alpha, \beta)}{p(\theta, \alpha, \beta)} \cdot p(\theta \mid \alpha)$$

$$= p(\mathbf{w} \mid \mathbf{z}, \theta, \alpha, \beta) \cdot p(\mathbf{z} \mid \theta, \alpha, \beta) \cdot p(\theta \mid \alpha)$$

Since z only depends on θ and w only depends on z and β,

$$= p(\mathbf{w} \mid \mathbf{z}, \theta) \cdot p(\mathbf{z} \mid \theta) \cdot p(\theta \mid \alpha)$$

Thus, it is shown that for each n of N:

$$p(\theta, \mathbf{z}, \mathbf{w} \mid \alpha, \beta) = p(\theta \mid \alpha) \prod_{n=1}^{N} p(z_n \mid \theta) p(w_n \mid z_n, \beta) \quad \blacksquare$$

where $p(z_n \mid \theta)$ is simply θ_i for the unique i such that $z_n^i = 1$. Integrating over θ and summing over z, we obtain the marginal distribution of a document:

$$p(\mathbf{w} \mid \alpha, \beta) = \int p(\theta \mid \alpha) \left(\prod_{n=1}^{N} \sum_{z_n} p(z_n \mid \theta) p(w_n \mid z_n, \beta) \right) d\theta.$$

Finally, taking the product of the marginal probabilities of single documents, we obtain the probability of a corpus:

$$p(D \mid \alpha, \beta) = \prod_{d=1}^{M} \int p(\theta_d \mid \alpha) \left(\prod_{n=1}^{N_d} \sum_{z_{dn}} p(z_{dn} \mid \theta_d) p(w_{dn} \mid z_{dn}, \beta) \right) d\theta_d.$$

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